

Surveillance Drone

6-DOF quadrotor, GPS/ENU waypoint patrol with geofence + RTH, ground-frame intruder detection, tracking, alerts

AT A GLANCE

12/12 (mean err 1.48 m)

Waypoints reached

0 / 1

Fence violations / clamps

7/7

Intruders detected

97.7 / 99.5 %

Detection recall / precision

0.001

False positives per frame

12 (12 / 0)

Alerts (true / false)

1.23 / 1.60 s

Alert latency mean / max

13/13

Unit tests

VERDICT

SIM PASS

TESTS 13/13

0 FENCE BREACHES

DECLARED SUBSTITUTION

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1. Objective and brief mapping

I simulated a surveillance quadrotor end to end in pure Python: a simplified 6-DOF rigid-body model with four lagged rotors and a cascaded position -> attitude -> rate controller; GPS waypoints in WGS84 lat/lon/alt converted through ECEF into a local ENU frame anchored at the take-off point (Ghaziabad); a polygon geofence with margin and altitude ceiling that clamps illegal targets and triggers return-to-home on a breach; a momentum-theory battery model with a low-SoC RTH; a patrol state machine (take-off, patrol, hold, RTH, land); a top-down renderer of the compound (buildings, roads, trees) with walking people and driving vehicles seen through a nadir camera whose footprint follows the altitude; OpenCV detection in a ground-fixed frame (MOG2 + contours, reference-map differencing, HOG-SVM shape classifier), a centroid tracker with IDs and an alert engine with intrusion zones, dwell rule, JSON / PNG / CSV outputs and statistics.

Every number below was measured in this container by `sim/run_sim.py` (`sim/metrics.json`, `sim/alerts/`, `sim/flight_log.csv`, `sim/mission_events.log`) and the unittest suite.

2. Architecture

`src/drone/` is imported unchanged by the simulator, the tests, the ROS 2 node and the AirSim client.

```
Waypoint(lat, lon, alt) --GeoRef (WGS84->ECEF->ENU)--> ENU + home offset --Geofence.clamp--> Mission
TAKEOFF -> PATROL (carrot at cruise speed) -> HOLD -> ... -> RTH (patrol done | SoC < 25 % | fe
Mission.step: Quadrotor.control(target, yaw) -> 4 rotor thrusts -> Quadrotor.step (100 Hz) ; Ba
Camera.render(scene, pos): footprint = 2 h tan(fov/2), warp static raster + draw intruders, noise/
GroundFrameDetector: frame -> ground canvas (4 px/m) -> MOG2 -> motion blobs | frame vs refere
fuse (reference primary, motion = 'moving' flag / gated by HOG-SVM) -> world
CentroidTracker (4 m gate, 8-frame memory, IDs) -> AlertEngine (zones, dwell 1-1.5 s, cooldown 8 s
```

3. Implementation details

- **Geodesy:** WGS84 ellipsoid ($a = 6378137$ m, $f = 1/298.257$), closed-form LLA -> ECEF and an iterative ECEF -> LLA; ENU by the standard rotation at the home latitude. Round trips are exact to $1e-8$ deg (test); 0.001 deg of latitude at 28.67 N maps to 110.6 m north, 0.001 deg of longitude to 97.5 m east.
- **Quadrotor:** $m = 1.4$ kg, $I = \text{diag}(0.021, 0.021, 0.036)$ kg m², 0.225 m arms in X, 8 N per motor with a 50 ms lag, linear drag 0.25 N s/m. Controller: position PD (k_p 1.2, k_d 1.8, $v \leq 6$ m/s) -> desired acceleration -> thrust and roll/pitch (tilt ≤ 30 deg) -> attitude P (7) -> rate PI -> mixer. Semi-implicit Euler at 100 Hz.
- **Mission:** a carrot 6 m ahead along the line gives a steady 4 m/s cruise; acceptance radius 1.5 m; WP6 was placed 21 m outside the fence on purpose and is clamped toward the polygon centroid until it satisfies the 3 m margin (logged once per waypoint).
- **Camera / scene:** 320 x 240 nadir frames, 70 deg FOV -> a 39 x 29 m footprint at 28 m (8.2 px/m); a person is $\sim 5 \times 8$ px, a car $\sim 34 \times 16$ px. Per-frame gain jitter and $N(0, 3)$ noise. Seven intruders: four people (one loiters in zone A, one wanders outside every zone, one enters zone B at 70 s, one stands still at the north gate from 100 s) and three vehicles (two on the roads, one that leaves the road at 120 s and parks inside zone A).
- **Detection:** because the camera moves, MOG2 runs on a georeferenced ground canvas rather than on raw frames, and only on pixels the model has seen at least 8 times. The primary detector is gain-normalised differencing against the clean reference raster of the static compound (threshold 40, open 2×2 , close 3×3 , blobs 0.25-25 m²), which also finds stationary intruders. Motion-only blobs are accepted only if 0.3-8 m², fully inside the frame and classified person/vehicle at > 0.6 by the HOG-SVM; otherwise MOG2 just sets the 'moving' flag. The SVM (HOG 8 px cells on 32×32 + hue histogram + aspect/area, $C = 2$, RBF) is trained on crops extracted through this exact pipeline from single-intruder renderings (300 samples, held-out accuracy 100.0 %).
- **Tracker / alerts:** greedy nearest-neighbour association on velocity-predicted positions (4 m gate), confirmation after 2 hits, deletion after 8 misses, majority class vote. Alerts need a confirmed track inside a zone for the dwell time (1.0-1.5 s); an 8 s cooldown per (track, zone) re-alerts while the intruder stays.

4. Verification

`python3 -m unittest discover -s tests -v: 13 tests, all passing.`

Test group	What is checked	Result
Geodesy (3)	LLA/ECEF round trips at four sites, ENU axis scale and direction at Ghaziabad, geofence contains / margin / ceiling / clamp / boundary distance	pass
Quadrotor (4)	reaches a 3-D target to 0.3 m and hovers at m g, tilt and speed stay within limits on a 200 m dash, yaw tracks 1.2 rad, battery power law and SoC	pass
Mission (2)	3 legal + 1 illegal waypoint: legal ones reached within 1.5 m + 0.5 m, drone never crosses the fence margin, illegal one clamped, lands at home; low-SoC RTH fires and lands	pass
Perception (4)	HOG-SVM held-out ≥ 85 %, pixel->world mapping and footprint formula, detector hits 3 injected intruders in ≥ 80 % of 60 frames with ≤ 6 false positives and ≤ 6 tracks, alert fires 1.0 s (dwell) after zone entry and re-alerts after the cooldown	pass

KEY TAKEAWAY Two patrol loops over a 160 x 120 m compound: 12/12 GPS waypoints, 0 geofence breaches, all 7 intruders detected at 98 % recall, 12 alerts with 0 false and a worst latency of 1.6 s. AirSim / PX4-Gazebo / ROS 2 artefacts provided, not run here.

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5. Measured performance (sim/metrics.json)

Latency is measured from the first camera frame in which an intruder is both inside a zone and inside the footprint to the first alert of that visit, so it includes the dwell rule (1.0-1.5 s) and track confirmation (2 frames = 0.2 s); the values are the dwell time plus one or two frames, which is what the design says. A detection counts as a false positive only when no active intruder is within 3.5 m; the few that remain are motion-only blobs from MOG2 smearing at cruise speed, and none of them produced an alert, because alerts require a confirmed track dwelling inside a zone. The 29 missed intruder-frames belong almost entirely to the vehicle that parks inside zone A: for a few seconds it stands on the road/grass boundary where the gain-normalised difference to the reference is weak.

Area	Metric	Value
Mission	state / duration / distance flown	DONE / 454.4 s / 1083.4 m
Mission	waypoints reached / mean and max error at acceptance	12/12 / 1.477 m, 1.497 m
Mission	fence clamps (illegal WP6, once per pass) / violations / landing error	1 / 0 / 0.0 m
Battery	energy used / final SoC (74 Wh pack)	5.06 Wh / 93.2 %
Battery RTH test (SoCo = 26.2 %)	RTH trigger / landed at / landing error / final SoC	78.22 s / 303.9 s / 0.0 m / 21.6 %
Geofence test	target (200, 60, 80) clamped to / max x reached / stayed inside	[148.26, 60.0, 60.0] / 150.0 m / True
Detection	camera frames / intruder-frames in view / TP / missed	4144 / 1281 / 1252 / 29
Detection	recall / precision / FP per frame (by source)	97.7 % / 99.5 % / 0.0014 ({'reference': 0, 'motion': 6})
Detection	localisation error / class accuracy on TPs / ms per frame	0.15 m / 97.7 % / 18.1 ms
Tracking	tracks created for 7 intruders (re-acquisitions after leaving the footprint)	25
Alerts	total / true / false / per minute / snapshots	12 / 12 / 0 / 1.58 / 12
Alerts	first-alert latency after in-view zone entry: mean / max / list (s)	1.23 / 1.6 / [1.6, 1.3, 1.0, 1.1, 1.0, 1.5, 1.1]

5b. Mission event timeline (sim/mission_events.log, first 9 events)

t (s)	Event	Detail
0.00	waypoint_outside_fence_clamped	WP6_outside: [175.0, 60.0, 28.0] -> [149.8, 60.0, 28.0]
5.64	takeoff_complete	
16.59	waypoint_reached	WP1
39.47	waypoint_reached	WP2
55.27	waypoint_reached	WP3
77.48	waypoint_reached	WP4
91.44	waypoint_reached	WP5
108.31	waypoint_reached	WP6_outside
132.63	waypoint_reached	WP1

6. Per-intruder detection and alert log

Intruder	Kind	Frames in view	Frames detected	Detection rate
0	person	343	343	100.0 %
1	person	44	44	100.0 %
2	person	128	128	100.0 %
3	person	174	174	100.0 %
4	vehicle	238	238	100.0 %
5	vehicle	195	195	100.0 %
6	vehicle	159	130	81.8 %

Each alert also exists as sim/alerts/alert_NNN.json and an annotated snapshot PNG; the GPS position of the intruder comes from the ENU -> LLA inverse of the geodesy module.

#	t (s)	Zone	Class (conf)	Track	lat, lon	Dwell (s)	True
1	36.2	A_east_compound	person (0.525)	4	28.6694873, 77.4550494	1.6	True
2	44.3	A_east_compound	person (0.525)	4	28.6695568, 77.4550901	9.7	True
3	114.5	A_east_compound	person (0.467)	9	28.6695819, 77.4549981	1.5	True
4	151.0	A_east_compound	person (0.722)	13	28.6695763, 77.4549666	1.6	True
5	151.4	A_east_compound	vehicle (0.676)	14	28.6695463, 77.4549983	1.6	True
6	157.1	A_east_compound	person (0.891)	15	28.6695127, 77.4550037	1.6	True
7	159.5	A_east_compound	vehicle (0.676)	14	28.6695466, 77.4549984	9.7	True

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Robotics & Automation
Codec Technologies India
Internship 2026

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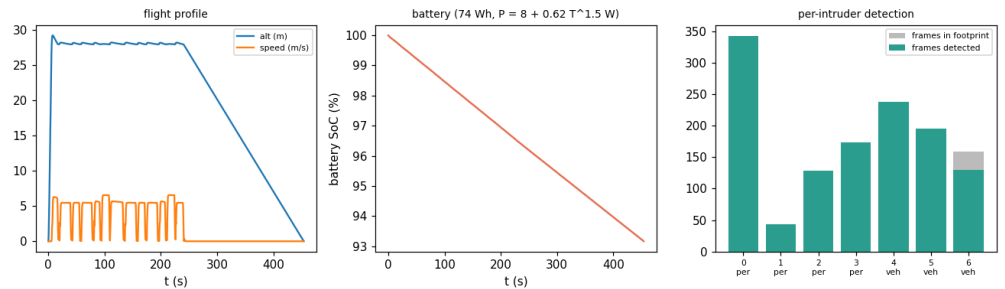
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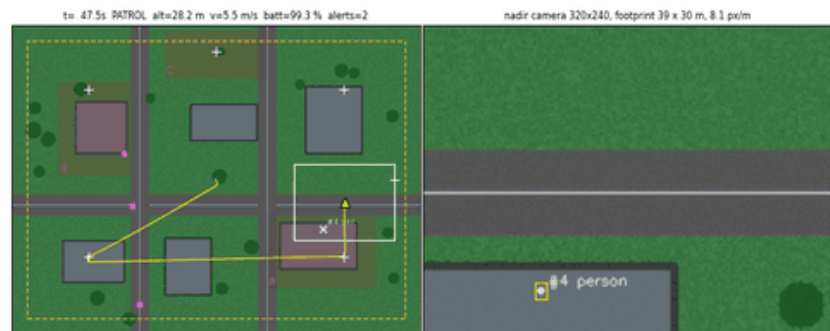
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#	t (s)	Zone	Class (conf)	Track	lat, lon	Dwell (s)	True
8	184.3	C_north_gate	person (0.951)	18	28.6701744, 77.4545167	1.1	True
9	191.0	B_west_store	person (0.745)	19	28.6699499, 77.4541815	1.6	True
10	199.1	B_west_store	person (0.745)	19	28.6699929, 77.4542371	9.7	True
11	203.8	C_north_gate	person (0.823)	21	28.6701737, 77.4545171	1.1	True
12	211.9	C_north_gate	person (0.823)	21	28.670174, 77.4545172	9.2	True

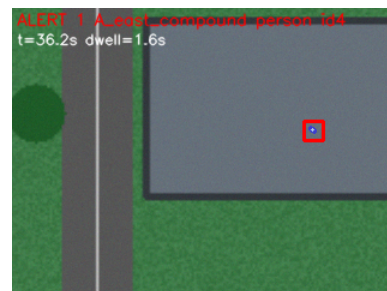
7. Figures



Flight profile (altitude, speed), battery state of charge, and per-intruder frames in view vs frames detected.



GIF frame: map with geofence (orange), intrusion zones (red), footprint (white), path (yellow), ground-truth intruders (magenta) and tracks (cyan); right: the nadir camera with detections and track IDs.



Annotated alert snapshot written by the alert engine (sim/alerts/snapshots/).

8. Declared substitutions / not run here

- **AirSim**: external/airsim_patrol.py drives an AirSim multirotor (takeoff, NED position setpoints from the same geofence-clamped ENU waypoints, bottom_center camera into the same detector/tracker/alert engine). The airsim package and Unreal are not available; the script prints a notice here.
- **PX4 + Gazebo + MAVROS**: external/px4_gazebo.launch.py starts PX4 SITL (gz_x500), MAVROS, the ros_gz camera bridge and the surveillance_drone node. Syntax-checked only.
- **ROS 2 Humble**: ros2_ws/src/surveillance_drone/ (package.xml, setup.py, setup.cfg, resource marker, launch file, node.py consuming /mavros/global_position/global, /mavros/battery and /camera/image_raw and publishing setpoints, detections and alerts). py_compile-checked only.
- **Video**: the full 906-frame patrol is sim/drone_patrol.mp4 (H.264, 1.1 MB); the GIF is an 80-frame, 40 s excerpt around the first alert to stay under 2 MB.

9. Learnings / next steps

Background subtraction from a moving camera only works once you decide what 'background' means: warping every frame into a ground-fixed canvas made MORG usable, but it still smears at cruise speed, and the honest fix was to make the reference-map detector primary and to let the HOG-SVM veto motion blobs. Training the classifier on crops that went through the same detector was worth more than any feature tweak. Next: replace the semi-implicit Euler integrator by RK4 and add wind gusts, use a homography from tracked ground features instead of the known pose for the canvas warp (as a real system must), and run the PX4 launch file in Gazebo.

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